IVABRADINE

Requires sinus rhythm

Phosphenes (visual)

CHEMICAL NAME 3-[3-({[(7S)-3,4-dimethoxybicyclo[4.2.0]octa-1,3,5-trien-7-yl]methyl}(methyl)amino)propyl]-7,8-dimethoxy-1,3,4,5-tetrahydro-2H-3-benzazepin-2-one (hydrochloride)

CLASS Selective sinus-node If (funny current) inhibitor

TRADE NAMES Procoralan · Corlanor · Coraxan

THERAPEUTIC USES / INDICATIONS

- Chronic heart failure (HFrEF ≤35%) in sinus rhythm with HR ≥70–75 bpm despite β-blocker (or β-blocker intolerant).
- Stable angina in sinus rhythm (HR ≥70) when β-blockers not tolerated.

ADMINISTRATION

- **PO** twice daily with meals.
- Only for patients in sinus rhythm.

DOSAGES

- **Start:** 5 mg twice daily (2.5 mg in elderly ≥75 yr).
- After 2 weeks: increase to 7.5 mg twice daily if resting HR >60; reduce if HR <50 or symptomatic bradycardia.
- Target resting HR 50–60 bpm.

PHARMACOKINETICS

- **Peak:** 1 h (delayed by food). Bioavailability ~40%.
- Metabolised by CYP3A4.
- Excreted in urine and faeces. **t½:** 2 h (effective 11 h).

PHARMACODYNAMICS

- Blocks **If (funny) channels** in SA node → slows diastolic depolarisation → ↓ heart rate.
- No effect on contractility, BP or AV conduction.
- Also acts on retinal Ih channels → phosphenes.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Phosphenes (transient luminous visual phenomena in dim light).
- Bradycardia.
- Headache, blurred vision, dizziness.

ADVERSE EFFECTS (SERIOUS)

- Atrial fibrillation (↑ risk).
- Symptomatic bradycardia, heart block.
- Uncontrolled BP.

CONTRAINDICATIONS & CAUTIONS

- Resting HR <70 before treatment; cardiogenic shock; acute MI.
- Sick sinus syndrome, SA block, 3rd-degree AV block; pacemaker-dependent.
- Unstable/acute heart failure; severe hypotension; severe hepatic impairment.
- Strong CYP3A4 inhibitors; verapamil, diltiazem; pregnancy.

KEY INTERACTIONS

- **Ketoconazole, clarithromycin, HIV protease inhibitors:** contraindicated.
- **Verapamil, diltiazem:** contraindicated.
- **QT-prolonging drugs:** avoid.
- **Grapefruit juice:** ↑ levels.

NURSING MANAGEMENT

- Check apical pulse and rhythm before dose; hold if HR <50.
- Monitor ECG for AF.
- Ask about visual disturbances.
- Monitor HF symptoms and weight.
- Advise contraception.

PATIENT EDUCATION

- Take with food twice daily.
- Flashes of light in dim conditions are common and usually settle; take care when driving at night.
- Report palpitations, slow pulse, dizziness.
- Avoid grapefruit juice.

KEY POINTS

- If channel blocker — lowers HR only.
- Phosphenes; bradycardia; AF.
- HFrEF in sinus rhythm with HR ≥70.
- Avoid with CYP3A4 inhibitors, verapamil, diltiazem.

SOURCES CHECKED **[363]** Ivabradine 2.5 mg film-coated Tablets (UK SmPC) **[364]** Corlanor, oral (US label) **[365]** ivabradine, oral (US label) *Full references p. 432–457*